

Energy Shocks and the Green Adoption Channel in a Small Open Economy

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Abstract

We add an extensive-margin brown/green durable adoption choice to the small open economy HANK model of [Auclert et al. \(2024\)](#). Households choose discretely whether to switch to a green durable that insulates them from fossil energy prices, paying a resource cost. We study brown energy *price* and *supply* shocks under the two fiscal responses studied in the recent literature: a retail price cap ([Langot et al., 2026](#)) and a Slutsky-compensating transfer indexed to pre-crisis energy consumption ([Bayer et al., 2026](#)). The price cap stabilises output but mechanically eliminates the green adoption margin, while the transfer delivers comparable stabilisation and leaves adoption intact. Shielding consumers from the relative price is therefore not neutral for the energy transition. This is our contribution and, we think, its natural boundary: the three closest papers—[Auclert et al. \(2024\)](#), [Bayer et al. \(2026\)](#) and [Langot et al. \(2026\)](#)—all hold the household’s energy technology fixed, and we add the one margin they omit rather than the two-country or labour-market machinery they contain. The transition loss is linear in the cap’s intensity under a price shock and convex under a supply shock. As a secondary observation, the *sign* of the cap’s private-welfare effect flips with the elasticity of energy supply, matching the sign each of [Langot et al. \(2026\)](#) and [Bayer et al. \(2026\)](#) obtains in its own closure; we read this as locating one axis of their contrast, not as nesting their models. The policy *ordering* is robust to calibration; the *magnitude* of the adoption channel, and even the sign of its output effect under a price shock, depend on the calibrated switching cost and taste scale, which we report and discuss.

1 Introduction

Three recent papers study fiscal responses to the 2022 European energy crisis in HANK frameworks. [Auclert et al. \(2024\)](#) study a small open economy and stress the negative externality when all importers subsidise simultaneously. [Bayer et al. \(2026\)](#) use a two-country HANK with *fixed* euro-area energy supply and find subsidies are beggar-thy-neighbour while Slutsky transfers are not. [Langot et al. \(2026\)](#) evaluate the French *bouclier tarifaire* assuming *perfectly elastic* energy supply and find the price cap performs well. All three treat the household’s energy technology as fixed.

We relax exactly that margin, and only that margin. Households can adopt a green durable that insulates them from the fossil price, at a resource cost. The energy shock raises the return to adoption, so the crisis itself can accelerate the transition—unless policy removes the price signal that drives it. This is the contribution: an endogenous green-adoption margin, absent from all three references. We deliberately do *not* try to nest the two-country spillovers of [Bayer et al. \(2026\)](#) or the labour-cost employment channel of [Langot et al. \(2026\)](#). A small open economy with an exogenous foreign block contains neither, and competing on that machinery is not the point; our single comparative advantage is the margin the literature leaves out, and we play only that.

Because the price-versus-supply distinction turns out to matter for the adoption margin, we do let the energy-supply elasticity vary: with $\varepsilon^E = \infty$ the economy is a price taker (elastic

supply, as in Langot); with ε^E finite the quantity is fixed and the world price clears the market (as in Bayer). Both closures are reported below. This is a modelling convenience for our own question, not a claim to reconcile the two papers: the supply elasticity is one dimension along which they differ, and not the one that carries their headline mechanisms (Section 5).

2 Model

2.1 Environment

The aggregate structure is Auclert et al. (2024): a small open economy with sticky wages, a CES consumption basket nesting energy against a home/foreign bundle, imported energy and foreign goods with sticky retail prices, and a mutual fund holding claims on domestic firms, importers and a share ζ^E of the energy endowment. We build the model literally on the ARS aggregate blocks and replace its representative, homothetic household with a heterogeneous-agent household that carries a discrete durable state; everything else in the aggregate economy is inherited unchanged.

2.2 Model at a glance: the maintained assumptions

Because the household block departs from ARS in several coupled ways, we list the maintained assumptions explicitly before deriving anything. A reader who accepts this list can reconstruct every result.

1. **One new margin.** The only structural addition to Auclert et al. (2024) is a discrete brown/green durable adoption choice. Conditional on the durable state, preferences are ARS's: the same CES energy nest (α_E, η_E) and the same home/foreign nest. There is no within-durable-type energy-share heterogeneity.
2. **Green insulates from the fossil price.** A green adopter buys energy at P_G^E , which in the baseline ($\rho_g = 0$) is constant and does *not* respond to the fossil shock. Adoption is therefore, in the baseline, full insulation—an upper bound on the channel.
3. **Green is domestic and resource-cheaper; brown is imported.** The energy an adopter uses comes from a domestic technology whose marginal cost is near zero once the durable is installed—photovoltaic electricity, or the ambient heat a heat pump moves per unit of electricity (its coefficient of performance)—whereas brown energy is imported. An adopter's switch is therefore a genuine reduction in the country's *fossil import bill*, not a domestic transfer, which is what turns the current-account entry of Section 2.8 into an accounting identity rather than an assumption. We do not model the technology that supplies the green energy (no green production sector, no green investment).
4. **The switching cost is a resource cost booked externally.** Adopting costs ψ_g units, booked in the current account as an import, by accounting necessity under ARS's markup/capitalised-profits convention (Section 2.8).
5. **Unit of account is the domestic good.** We set $p^{num} = p_H$ (Section 2.3). This is a measurement choice, orthogonal to the nominal anchor; it keeps the resource constraint independent of how the CPI weights brown and green energy.
6. **Energy supply is a single dial.** $\varepsilon^E = \infty$ gives the price-taking closure (exogenous world price, price shock); ε^E finite fixes the quantity and lets the world price clear (supply shock). The steady state is identical under both.
7. **Monetary baseline is the constant-real-rate rule** of Auclert et al. (2024). Under this rule the real exchange rate is pinned, so a domestic transfer has no terms-of-trade effect; we also report a Taylor rule.

8. **First order throughout.** All impulse responses are first-order (sequence-space Jacobian). The steady state is common to every experiment, since both policy instruments are zero at the steady state.

2.3 Unit of account

The household block is numeraire-generic: it reads a price of the unit of account relative to the CPI, p^{num} , a real return $1 + r^{num} = (1 + r)p_{-1}^{num}/p^{num}$, and after-tax labour income $atw^{num} = atw/p^{num}$, and returns asset holdings converted by $A^{CPI} = Ap^{num}$ before meeting CPI-denominated objects in asset market clearing and the current account. We set $p^{num} = p_H$, the domestic good. This makes the resource constraint independent of how the CPI weights brown and green energy, which matters once the two coexist (Section 2.7). Everything contractually written in CPI units—fiscal transfers, the ARS subsistence term, and the switching cost ψ_g —is divided by p^{num} on the way into the budget.

Two remarks make the convention transparent. First, the choice of numeraire is the choice of nominal anchor, not of deflator: the Fisher equation is written once in CPI terms ($1 + r^{ante} = (1 + i)/(1 + \pi)$), and $p^{num} = p_H$ then delivers, after the relative-price conversion, a household discount factor deflated by *core* (domestic-good) inflation, $1 + r^{num} = (1 + i_{-1})/(1 + \pi^H)$. Second, output y is a physical quantity of the domestic good and is therefore already expressed in numeraire units—no re-deflation is needed to read it as a real quantity; CPI-deflated GDP would be $p_{H/P}y$, and the two differ only by the (energy-driven) movement in $p_{H/P}$.

2.4 The durable state

Each household carries two binary durable indicators: the technology it operates this period, $g \in \{B, G\}$, and the technology it held last period, $g_- \in \{B, G\}$. The switching cost ψ_g is incurred exactly in the transition $g_- = B \rightarrow g = G$ —a household that is green now but was brown last period. Brown is absorbing; a green durable breaks down to brown at rate δ_g , so a permanent green share is sustained by a flow of switchers. These are *two* states, each taking two values; *computationally* we collapse the pair (g, g_-) into a single four-valued index, because the sequence-space household block carries one durable dimension. Of the four combinations, $(g, g_-) = (B, G)$ —brown now, green last period—never occurs on the equilibrium path: no household abandons the cheaper technology, and green is lost only through breakdown, which is already recorded as $g_- = B$. It therefore carries zero mass and the effective state is three-point. This is an artefact of the collapse, not an economic assumption: the model has two binary technology states, not one four-way categorical state. Stages, in forward order, are [**dep**, **prod**, **durables**, **consav**]: breakdown, idiosyncratic productivity, the discrete adoption choice (a logit with taste scale σ_ε), and the EGM consumption–savings step. The adoption stage uses the discrete-continuous structure of Iskhakov et al. (2017).

Three subtleties are worth making explicit, because the calibration and the sign results depend on them. (i) The choice compares continuation values across durable states; the fiscal transfer enters cash-on-hand *identically* across durable states, so it shifts adoption only through the curvature of the value function, which is why the transfer barely moves the green share. (ii) The asymmetry “brown absorbing, green depreciating at δ_g ” means green ownership requires *active* re-adoption each time the durable breaks down, so a permanent green share is sustained by a flow of switchers $D^{sw} = \delta_g D^G$ in steady state. (iii) The logit taste scale σ_ε smooths the choice: at exact indifference the model assigns a structurally fixed switching probability (not one half), a consequence of the asymmetric choice sets across states. The magnitude of the adoption response is governed jointly by ψ_g and σ_ε ; see Section 6.

2.5 Energy prices

We distinguish three prices, all relative to the CPI:

$$\begin{aligned}
 p^E & \text{ wholesale/retail market price of brown energy, before any subsidy} \\
 P_B^E &= (1 - \tau^E) p^E + \tau^E \bar{p}^E && \text{price paid by a brown household} \\
 P_G^E &= \varrho \bar{p}^E (P_B^E / \bar{P}_B^E)^{\rho_g} && \text{price paid by a green adopter}
 \end{aligned}$$

with ϱ the steady-state green/brown ratio and ρ_g the pass-through of the fossil shock to the green price. In the baseline $\rho_g = 0$, so P_G^E is constant: adopters are fully insulated.

2.6 How the durable enters the budget

Households of different durable types face different energy prices, hence different *consumption bundle prices*. We give each type its own exact CES cost-of-living index, reusing ARS's own energy-nest parameters (α_E, η_E) :

$$p_d^E = P_B^E + \mathbb{I}[d \text{ green}] (P_G^E - P_B^E), \quad p_d = \left[1 + \alpha_E ((p_d^E)^{1-\eta_E} - (P_B^E)^{1-\eta_E}) \right]^{\frac{1}{1-\eta_E}},$$

and the budget constraint is $(p_d/p^{num})c + a' = \text{coh}(d)$. The second expression is exact given the CES identity enforced by the equilibrium block, so brown households have $p_d \equiv 1$ at every date. With type-specific prices, aggregate energy demand is $\sum_d c_E(d)$, not the CES demand evaluated at the aggregate index—the two differ by Jensen's inequality (0.5% here)—so aggregate c_H , c_F , c_E are built from the household block's own heterogeneous-agent outputs.

2.7 What the CPI measures

With two energy prices in the economy, the energy leg of the CPI could be anchored on the brown price alone or on the share-weighted index $[(1 - D^G)(P_B^E)^{1-\eta_E} + D^G(P_G^E)^{1-\eta_E}]$. We anchor on the brown price: it is what statistical agencies do (the energy basket is not reweighted quarterly for technology adoption), it preserves $p_d \equiv 1$ for brown households exactly, and feeding D^G back into the CPI would close a DAG cycle requiring the CPI to be promoted to a model unknown. Because the unit of account is the domestic good, this choice does not touch the resource constraint; it affects only the measured deflator and, through it, the monetary rule. The gap between the share-weighted and brown-anchored CPI has closed form

$$\phi^{1-\eta_E} = 1 + \alpha_E D^G \left[(P_G^E)^{1-\eta_E} - (P_B^E)^{1-\eta_E} \right],$$

computed ex post outside the model. On the price shock without policy it reaches 0.14 annualised percentage points, about 1.5% of the amplitude of measured CPI inflation (0.23 pp, 2.3%, on the supply shock; Figure 1).

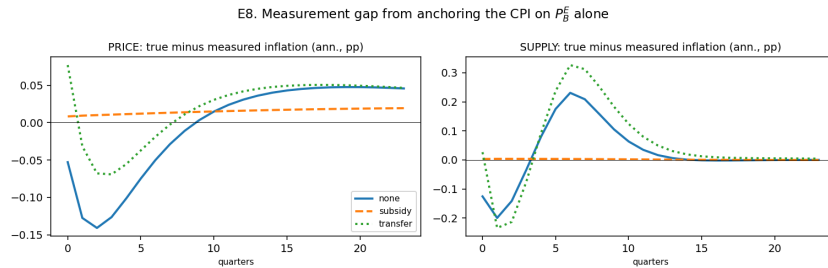


Figure 1: Measurement gap from anchoring the CPI on P_B^E alone: true minus measured annualised inflation, computed ex post.

2.8 Balance of payments

Both objects below are required, not optional. Once households pay a switching cost and pay two different energy prices, their summed budget constraints carry resource flows that must have an aggregate counterpart, or Walras' law fails and the current account and asset market do not clear. The two terms are those counterparts.

We implement two internally consistent bookings of these flows, selectable by a single switch, and report both; the choice has a first-order effect on the adoption channel's output sign (Section 4.2), so it is a result, not a nuisance.

Import booking (baseline). The **switching cost** is booked as an import, $\text{imports}_{dur} = \psi_g D^{sw}$: booking it as domestic demand here creates a marginal dividend owned by nobody, since with $\mu_{ss} > 1$ and capitalised profits extra demand for the home good is not competitively supplied. The **energy saving** of adopters, $(P_B^E - P_G^E) C_E^G$, is booked as a real import saving—cheaper energy for adopters is treated as a genuine external resource saving, giving the windfall an external counterpart consistent with asset-market clearing. All energy, brown and green, is imported at the world brown price and the green gap is netted off, so green energy remains partly an import.

Domestic booking (green as a domestic industry). We also model green explicitly as a domestic sector. Brown energy is imported at the world price; green energy is supplied by a domestic, competitive, near-zero-marginal-cost technology—photovoltaic electricity, or the ambient heat a heat pump moves per unit of electricity—and the green durable is installed by the same domestic sector at zero profit. Both flows, the green energy bill $P_G^E C_E^G$ and the installation $\psi_g D^{sw}$, are domestic value added, rebated to households lump-sum; only *brown* energy is imported and only brown clears against the world supply. This resolves the phantom-dividend problem that motivates the import convention (the rebate is a competitive, zero-profit sector, so there is no dividend owned by nobody), and the steady state clears to machine precision. The two bookings are genuinely non-equivalent: the import booking leaves green partly in the import bill, the domestic booking removes it entirely and credits the economy with $P_G^E C_E^G$ of domestic value added, a difference of about 0.16% of GDP at the steady state. Because green production is modelled as costless value added (no green investment, no factor use), the domestic booking is an *upper bound* on the adoption benefit (Section 6); the robust object is the sign of the channel, which differs across the two bookings.

2.9 Policy instruments

Both instruments are the extreme (full-compensation) versions, as in Bayer et al. (2026).

$$\text{Price cap: } P_{B,t}^E = (1 - \tau^E) p_t^E + \tau^E \bar{p}^E, \quad \tau^E = 1 \Rightarrow \text{retail price fixed at pre-crisis level} \quad (1)$$

$$\text{Slutsky transfer: } tr_{it} = \iota (p_t^E - \bar{p}^E) \bar{c}_{E,i}^B, \quad \iota = 1 \Rightarrow \text{full compensation on pre-crisis brown use} \quad (2)$$

Both are debt financed and repaid through the distortionary labour tax. The transfer base $\bar{c}_{E,i}^B$ is household i 's steady-state *brown* energy demand, not its total energy demand: the shock raises only the brown price (P_G^E is fixed under $\rho_g = 0$), so the exact Slutsky compensation is indexed on brown consumption alone; indexing on total energy would over-compensate the 5% steady-state green mass, whose price does not rise. Crucially the transfer is a genuine *lump sum*: the base is the pre-crisis brown bill and it is paid identically whatever durable the household currently holds, so it leaves the current brown–green relative price intact. Making it durable-state-contingent instead—paying only households currently brown, and so nothing to a household

that switches—would tax the switching margin and spuriously shut the adoption channel down; it is precisely the lump-sum indexing that lets the transfer preserve adoption (Section 4.4) where the cap eliminates it. Aggregated over the stationary distribution the base equals the economy’s pre-crisis brown energy bill, so the government’s outlay matches what households receive and the current account clears.

The cap’s fiscal base is booking-dependent. Only the brown price is capped, so under the domestic booking, where green is a domestic industry, the subsidy is indexed on brown consumption alone; under the import booking, where the green part of the subsidy has an import counterpart, it is indexed on total energy—the same brown/total distinction as the transfer, reflecting the 0.16%-of-GDP green import content that separates the two bookings (Section 2.8). The cap operates directly on P_B^E , exactly the price entering the brown–green gap, and therefore compresses the adoption incentive.

3 Calibration

Parameter	Value	Source / target
α_E energy expenditure share	0.04	Auclert et al. (2024)
η_E energy substitution	0.10	Auclert et al. (2024)
r, μ_{ss}	0.01, 1.03	Auclert et al. (2024)
β heterogeneity (3 types)	spread 0.06	fixed; implied MPC ≈ 0.30
δ_g green breakdown rate	0.05	assumption (5 yr life)
$\varrho = P_G^E / P_B^E$ steady-state gap	0.80	heat-pump / EV delivered-cost ratio
ψ_g switching cost	0.538	targets $D_{ss}^G = 0.05$
σ_ε taste scale	0.05	calibrated jointly with ψ_g
ρ_g pass-through to green price	0	upper bound on the channel

Table 1: Calibration. The steady state is identical across all experiments. ψ_g and σ_ε are calibrated jointly to the level target $D_{ss}^G = 0.05$; σ_ε is not separately pinned down by an adoption-elasticity moment (Section 6). The gross markup carries ARS’s home-endowment normalisation $\mu_{ss} \leftarrow \mu_{ss}(1 - \zeta^E \alpha_E)$, so the value entering firm profits is ≈ 1.016 ; 1.03 is the pre-normalisation primitive.

Robustness to the green–brown cost ratio ϱ . The one durable-side parameter with a direct empirical counterpart is the steady-state green/brown cost ratio ϱ . We anchor it at 0.80 from the relative cost of delivered energy service under adoption: a heat pump with a coefficient of performance near three delivers heat from roughly a third of the fuel-equivalent input, which at 2019–21 electricity-to-gas price ratios places the delivered green cost around 0.7–0.8 of brown; the per-kilometre cost of an electric against a comparable internal-combustion vehicle is in the same range. Table 2 re-solves the model across $\varrho \in \{0.6, 0.7, 0.8\}$, re-calibrating ψ_g each time to hold $D_{ss}^G = 0.05$. The *ordering* is invariant: the cap collapses the peak green-share response to essentially zero at every ϱ , the transfer leaves it at its no-policy value, the cap cushions cumulative output and the transfer boosts it. Only magnitudes move, and modestly (the no-policy peak green share ranges over 8.4–9.2 pp). The limiting case $\varrho = 1$ —green no cheaper than brown—is degenerate for the level target: no positive switching cost can sustain a 5% green steady state when adoption confers no cost advantage, so adoption collapses, which is exactly the sanity check one wants.

ϱ	ψ_g (re-solved)	no policy peak D^G	cap peak D^G	transfer peak D^G
0.60	0.714	8.39	0.03	8.53
0.70	0.624	8.81	0.02	8.95
0.80	0.538	9.18	0.01	9.32

Table 2: Robustness to the green–brown cost ratio ϱ (price shock; peak green-share response, pp over 24 quarters; ψ_g re-solved each row to $D_{ss}^G = 0.05$). Cumulative output moves monotonically but the ordering does not: no policy $-27.9 \rightarrow -30.1$, cap $-21.1 \rightarrow -20.9$, transfer $+14.9 \rightarrow +12.7$ as ϱ rises from 0.6 to 0.8.

4 Experiments and results

The *price* shock is a 100 log-point rise in the world energy price with a 16-quarter half-life, as in [Auclert et al. \(2024\)](#). The *supply* shock is a 10% cut in available energy for 6 quarters, after which the world price clears the market endogenously. Impulse responses are reported as $100 \times$ the level deviation from steady state. Variables whose steady state is one (prices, w , n , and the population shares D^G , D^{sw}) read directly as percent or percentage-point deviations, and C (steady state ≈ 0.997) is within rounding of a percent; y and gdp have steady state ≈ 0.985 , so their entries are level deviations $\times 100$, within about 1.5% of the exact percentage. Energy quantities are a share of the energy steady state (≈ 0.040).

Shock	Policy	$y(0)\%$	$\sum y$	$\pi(0)$ ann.	peak D^G pp	fiscal cost	adoption $\rightarrow \sum y$
price	none	−1.12	−30.1	+9.14	9.18	0.00	−1.60
	subsidy	−0.00	−20.9	−1.53	0.01	53.82	+0.36
	transfer	+1.03	+9.7	+10.94	9.46	49.23	−4.65
supply	none	−0.71	−8.2	+10.06	2.33	0.00	+0.38
	subsidy	−0.74	−46.8	−0.96	0.01	26.08	+0.18
	transfer	+1.16	+1.9	+14.13	3.35	24.00	+1.67

Table 3: Cumulative sums over 24 quarters. The last column is the contribution of the adoption margin: the difference between the model with the margin open and the same model with it shut.

4.1 The crisis moves the transition, in a shock-dependent direction

With no policy, the green share rises by up to 9.2 percentage points under the price shock (from 5.0% to 14.2% of households) and 2.3 pp under the supply shock (Figure 2). The output effect of the adoption margin, however, is not one-signed. Under the **supply** shock it is expansionary (+0.38 points of cumulative output with no policy, +1.67 under the transfer): adopters escape the energy bill and their real income falls less. Under the **price** shock it is contractionary (−1.60 points with no policy, −4.65 under the transfer): the switching cost is booked as an import in exactly the periods household income is falling hardest (Section 2.8), and at the calibrated ψ_g this resource cost outweighs the energy-bill saving. The sign of the adoption channel’s contribution to output is therefore not a general property of the model—it depends on the calibrated switching cost, on which shock is driving adoption, and (Section 6) on the accounting of ψ_g .

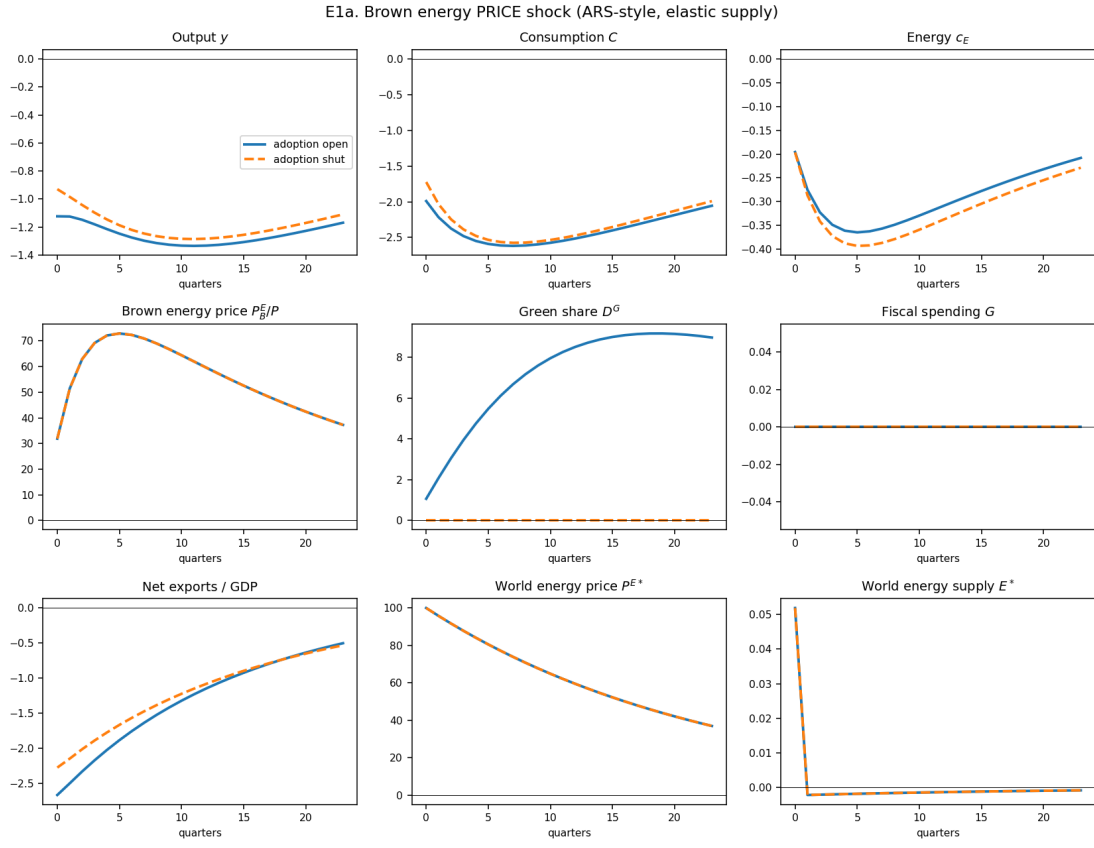


Figure 2: Brown energy *price* shock (elastic supply), adoption open (solid) vs. shut (dashed), no policy.

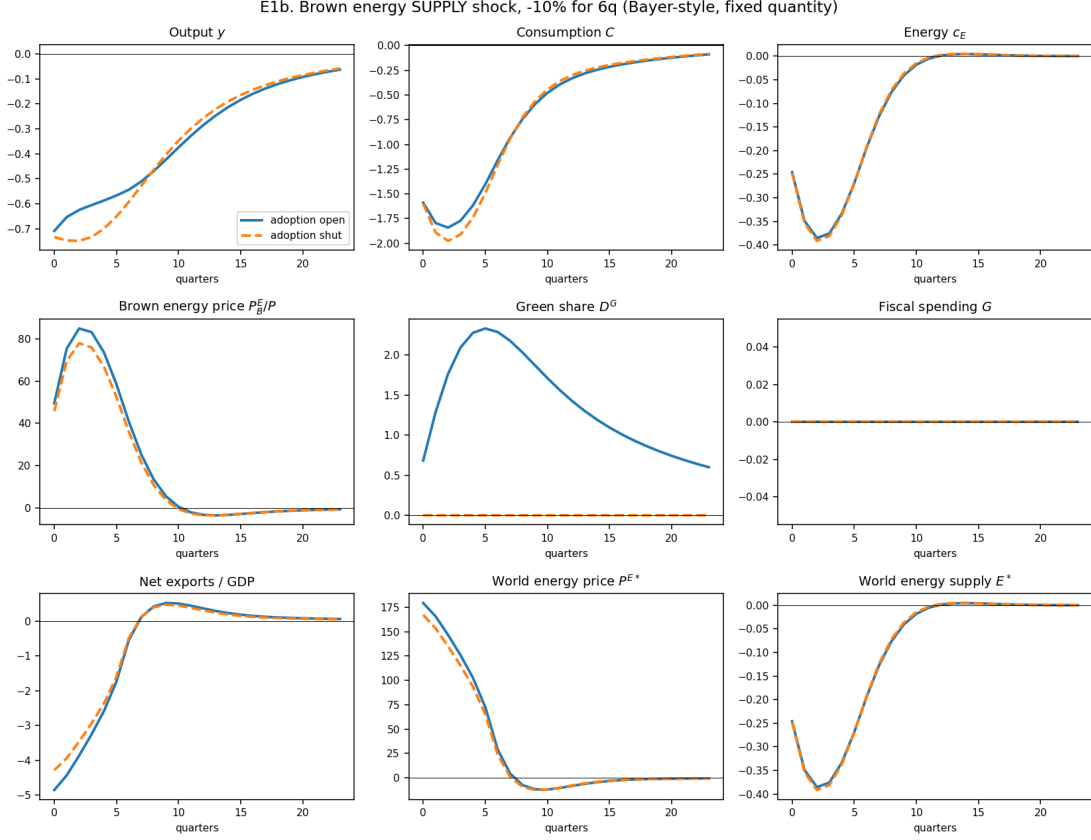


Figure 3: Brown energy *supply* shock (-10% for 6 quarters, fixed quantity), adoption open (solid) vs. shut (dashed), no policy.

4.2 A sign condition for the adoption channel

The shock-dependence of the adoption channel's output contribution is not accidental; it follows from a race between two resource flows, which we can characterise and then verify. Write the contribution to cumulative output over horizon H as $\Delta_H \equiv \sum_{t < H} (y_t^{\text{adopt}} - y_t^{\text{no-adopt}})$. Adoption moves two things in the resource constraint (Section 2.8):

- a **switching cost** $\psi_g D_t^{sw}$, booked as an import—a resource *outflow*, hence contractionary—which is *front-loaded* (paid once, up front, by each switcher) and scales with the adoption *volume* D^{sw} ;
- a **flow energy saving** $(P_B^E - P_G^E) C_{E,t}^G$, booked as reduced imports—a resource *inflow*, hence expansionary—which accrues *gradually* over the life of the green durable.

The sign of Δ_H is the sign of the second flow, cumulated over the window, minus the first. Two comparative statics follow.

Persistence (price closure). A more persistent brown-price shock raises the return to switching and therefore the adoption volume D^{sw} steeply (peak D^G rises from 1 to 22 pp as the half-life goes from 2 to 256 quarters). The front-loaded import cost thus grows faster than the windowed saving, so Δ_H is *decreasing* in persistence and crosses zero at a finite half-life—about 11 quarters in our calibration (Figure 4). Short-lived shocks trigger little adoption whose quick saving dominates ($\Delta_H > 0$); persistent shocks trigger a large adoption wave whose front-loaded cost dominates within the window ($\Delta_H < 0$). This is the opposite of the naive intuition that more

persistence, by raising the per-switcher saving, should make adoption more expansionary: the per-switcher calculus is dominated by the response of the switcher *count*.

Closure. Under the fixed-quantity (supply) closure a third, expansionary term appears that is absent under the price closure: adoption lowers fossil demand against a fixed supply, so the market-clearing price falls, relaxing the scarcity for *all* households. This shifts Δ_H upward, which is why the supply shock’s adoption contribution is positive (+0.38) at a persistence where the price shock’s is negative.

Booking. Both statics are conditional on the import booking of the adoption flows. We solve the model under the alternative domestic booking of Section 2.8—green energy and green installation as competitive, zero-profit domestic value added rebated to households, with only brown energy imported—which resolves the phantom-dividend problem and clears to machine precision. Under this booking neither the switching cost nor the green energy bill is a leakage, and the adoption channel’s contribution to cumulative output (Δ_{24}) flips sign:

Δ_{24} , adoption output contribution	import booking	domestic booking
price shock	−1.60	+35.98
supply shock	+0.38	+5.96

Table 4: Adoption channel contribution to cumulative output (Δ_{24}), import vs. domestic booking (Section 2.8). *None* policy.

The adoption wave itself is essentially unchanged across bookings (peak D^G 9.18 $\rightarrow$ 9.46 pp on the price shock, 2.33 $\rightarrow$ 1.34 on the supply shock): only the booking of its resource flows moves, not the margin. The domestic-booking magnitudes are an *upper bound*—green value added is modelled as costless, maximally stimulative—but the *sign reversal* is robust and confirms the characterisation: the sign of the adoption channel is a property of how its resource cost is booked, not of the adoption margin itself. Finally, the marginal switcher’s steady-state indifference, $\psi_g \approx (\bar{P}_B^E - \bar{P}_G^E) \bar{C}_E^G / (r + \delta_g)$, ties the threshold to the cost ratio ϱ and reproduces the direction of the $\psi_g(\varrho)$ mapping in Table 2. A closed-form threshold via the full Keynesian-cross algebra of the adoption block is left for the next draft; the characterisation above is verified numerically in Figure 4.

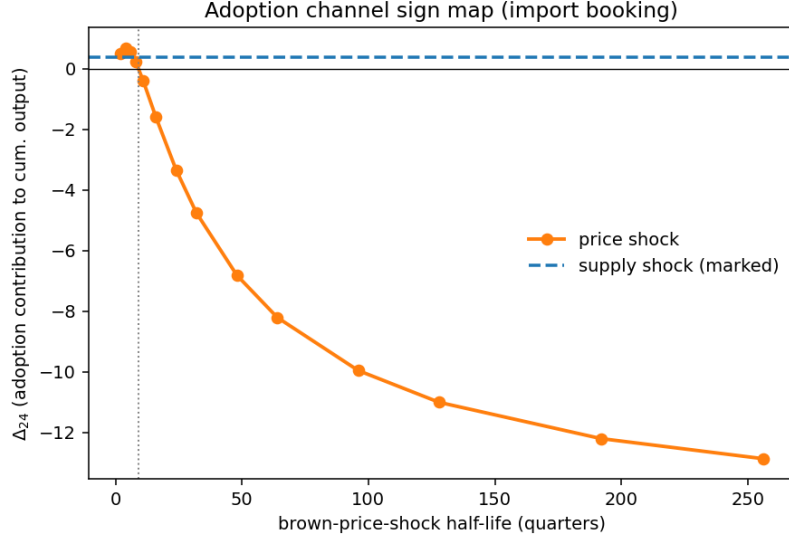


Figure 4: Adoption’s contribution to cumulative output, Δ_{24} , against the brown-price-shock half-life (price closure, orange), with the supply-shock value marked (blue). The contribution is expansionary for transitory shocks, crosses zero near an 11-quarter half-life, and turns increasingly contractionary as the shock—and hence the adoption wave whose front-loaded switching cost it imports—becomes more persistent.

4.3 The price cap eliminates the adoption margin

Under the cap, the peak green share response falls to essentially zero under both shocks (from +9.18 to +0.01 pp on the price shock, from +2.33 to +0.01 on the supply shock): the adoption channel is not attenuated, it is switched off. With the margin shut its contribution to cumulative output collapses toward zero accordingly. The mechanism is direct: the cap holds P_B^E at its pre-crisis level, and P_B^E is precisely what determines the return to adoption.

E2. Fiscal response to the PRICE shock (adoption margin open)

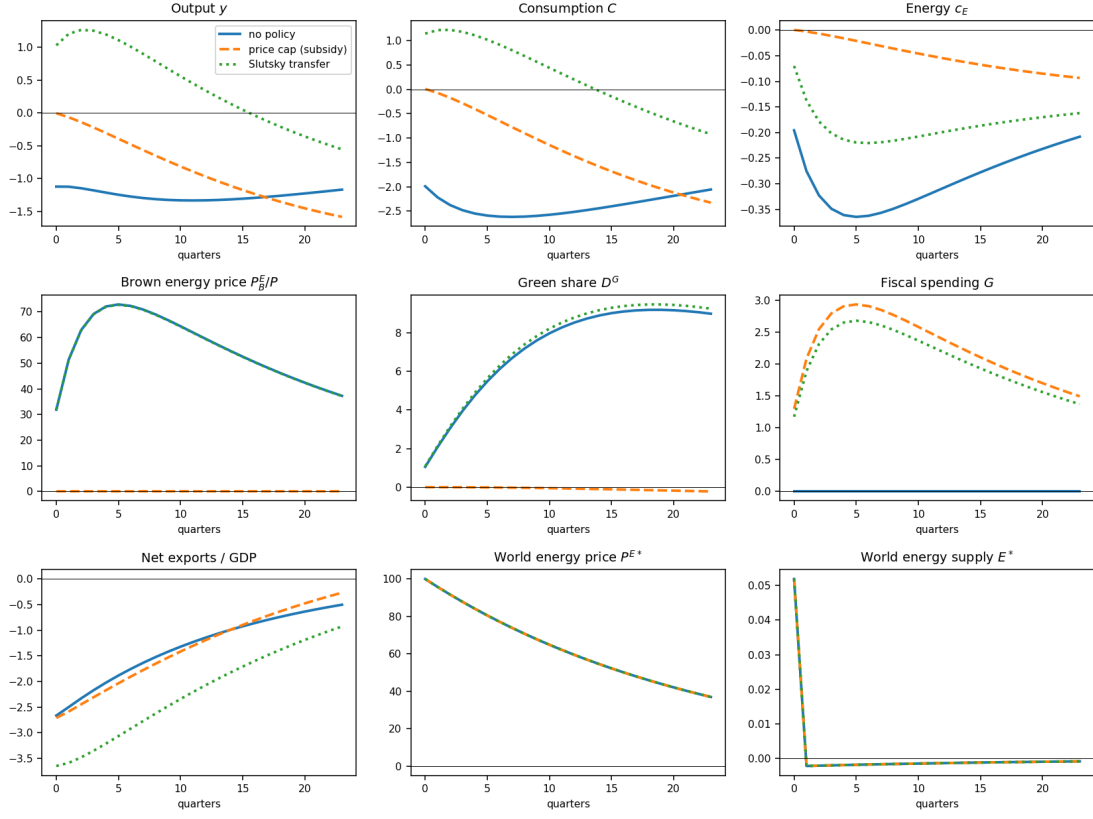


Figure 5: Fiscal response to the *price* shock, adoption open: no policy (solid), price cap (dashed), Slutsky transfer (dotted).

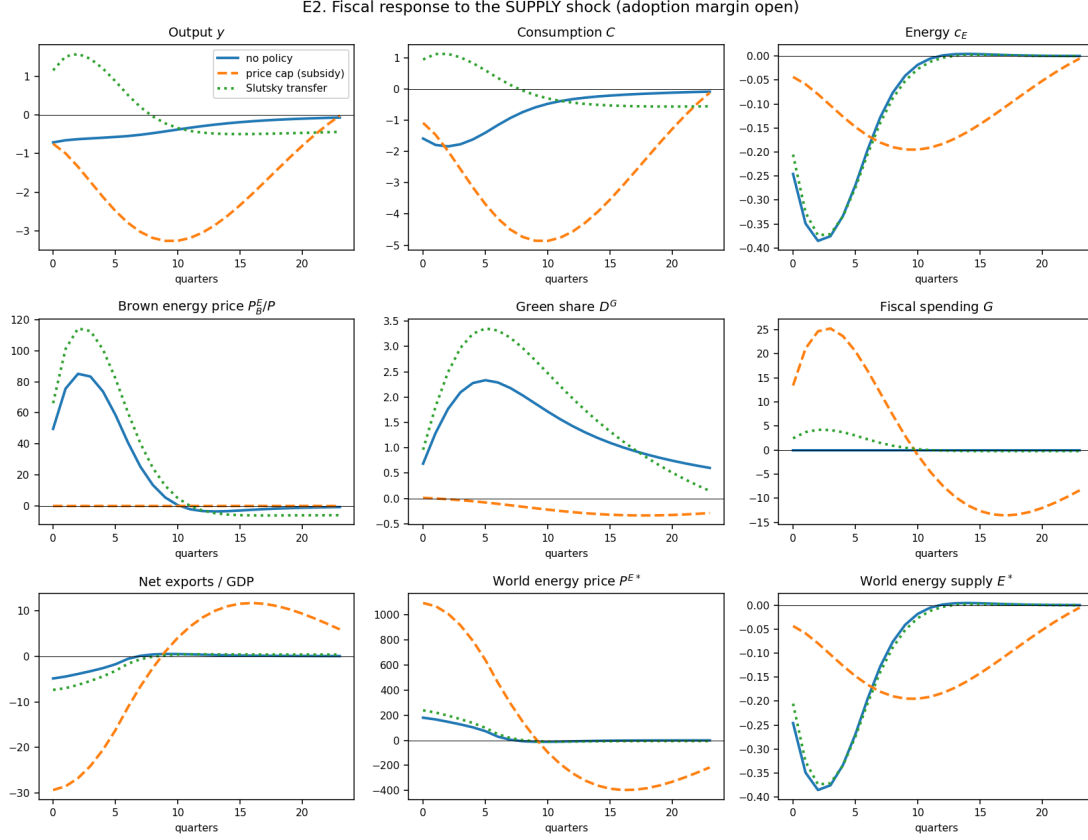


Figure 6: Fiscal response to the *supply* shock, adoption open: no policy (solid), price cap (dashed), Slutsky transfer (dotted).

4.4 The transfer preserves adoption, at a cost to output

The Slutsky transfer leaves the green share response essentially unchanged (9.46 vs. 9.18 pp under the price shock, 3.35 vs. 2.33 under the supply shock) while delivering more output stabilisation than the cap, at similar fiscal cost: subsidies destroy the transition margin, transfers do not. This preservation is a direct consequence of the transfer's lump-sum indexing (Section 2.8): because it does not condition on the household's current durable, switching forfeits nothing and the relative-price incentive to adopt is left intact. Leaving adoption intact is not free for output under the price shock, however: the transfer's cumulative-output gain is *smaller* with adoption open than shut (-4.65 points from the margin, Table 3), for the same resource-cost reason as above.

4.5 Fixed supply makes the cap destructive

Under the supply shock the cap yields cumulative output of -46.9 against -8.2 with no policy: with inelastic supply, capping the price prevents substitution away from energy and pushes the economy towards Leontief. This survives the addition of an adoption margin. It is also precisely the configuration in which [Langot et al. \(2026\)](#), assuming elastic supply, find the cap effective: our elastic-supply column agrees with them (-20.9 vs. -30.1 for cap vs. no policy under the price shock).

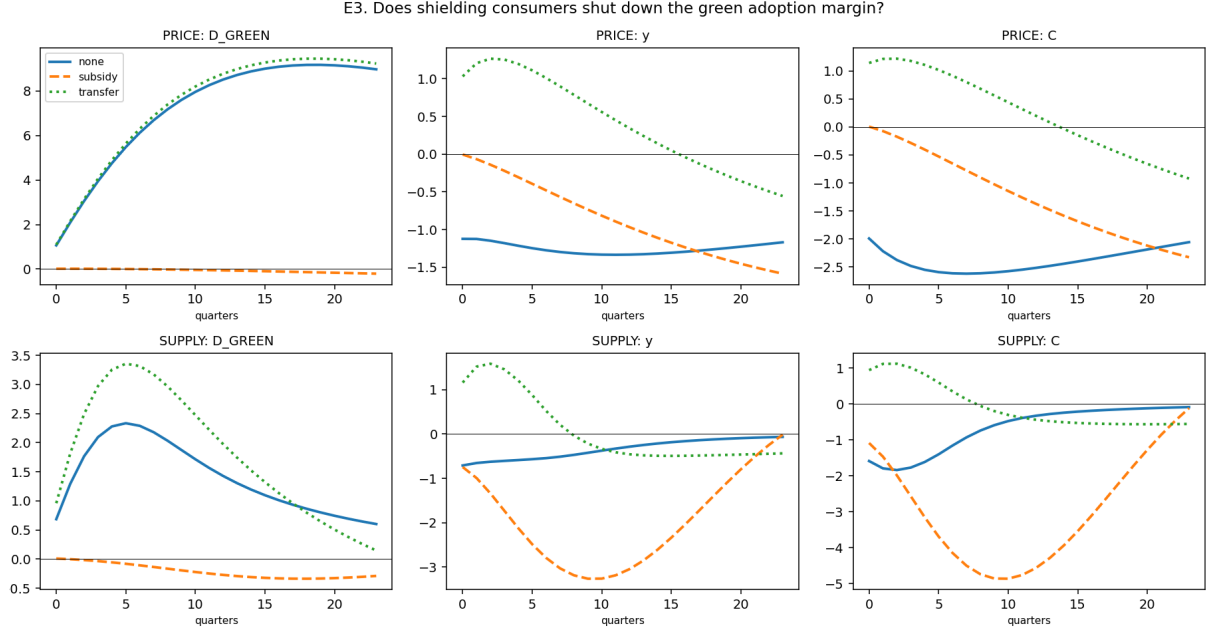


Figure 7: Policy and the adoption margin. Under both shocks the price cap (dashed) flattens the green share response to zero; the transfer (dotted) leaves it intact.

4.6 The cap is a dial, not a switch

Table 3 used $\tau^E = 1$, a complete price freeze. Table 5 sweeps $\tau^E \in [0, 1]$. Under the *price* shock the adoption loss is almost exactly linear in τ^E (half a cap costs a bit under half the transition: 4.53 against 9.18 pp). Under the *supply* shock it is convex: small caps barely touch adoption and the collapse is concentrated near the full freeze. A partial shield is therefore much less damaging to the transition when supply is inelastic—but, as the next result shows, much more damaging to everything else.

Shock	Policy	peak D^G pp	$\sum y$	$\pi(0)$	$\sum c_E$	fiscal	CEV %
price	cap $\tau^E = 0.00$	9.178	-30.06	9.14	-6.99	0.00	-1.3000
	cap $\tau^E = 0.25$	6.855	-27.78	6.50	-5.55	13.39	-1.1195
	cap $\tau^E = 0.50$	4.528	-25.50	3.84	-4.10	26.83	-0.9379
	cap $\tau^E = 0.75$	2.200	-23.22	1.17	-2.65	40.30	-0.7552
	cap $\tau^E = 1.00$	0.013	-20.93	-1.53	-1.20	53.82	-0.5714
	transfer	9.456	9.74	10.94	-4.45	49.23	-0.2513
supply	cap $\tau^E = 0.00$	2.333	-8.16	10.06	-2.40	0.00	-0.5863
	cap $\tau^E = 0.25$	2.211	-10.60	9.28	-2.40	5.96	-0.6267
	cap $\tau^E = 0.50$	1.987	-14.80	8.07	-2.40	15.34	-0.6971
	cap $\tau^E = 0.75$	1.501	-23.27	5.89	-2.43	30.89	-0.8425
	cap $\tau^E = 1.00$	0.010	-46.85	-0.96	-2.83	26.08	-1.2811
	transfer	3.355	1.92	14.13	-2.40	24.00	-0.1609

Table 5: Dose-response in the price-cap intensity τ^E , against the Slutsky transfer benchmark (full compensation, $\iota = 1$).

4.7 The welfare effect of the cap flips sign with the supply elasticity

This is the sharpest result in the paper. Under the price shock, tightening the cap *raises* welfare monotonically, from -1.30% to -0.57% ; under the supply shock it *lowers* welfare monotonically,

from -0.59% to -1.28% . The sign of the cap’s welfare effect therefore hinges on the supply elasticity. Langot et al. (2026), assuming elastic supply, find the French shield effective; Bayer et al. (2026), assuming fixed union-wide supply, find subsidies destructive. Our two closures reproduce the *sign* each obtains, which locates one source of the contrast in a single parameter. We do not claim to reconcile the two papers fully: Langot’s result also runs through a labour-cost channel and Bayer’s through cross-border spillovers in a monetary union, and our small open economy nests neither. The narrower point we establish is that, once the supply elasticity is fixed, the direction of the cap’s welfare effect is determined—so part of the empirical disagreement reduces to measuring that elasticity at the relevant horizon.

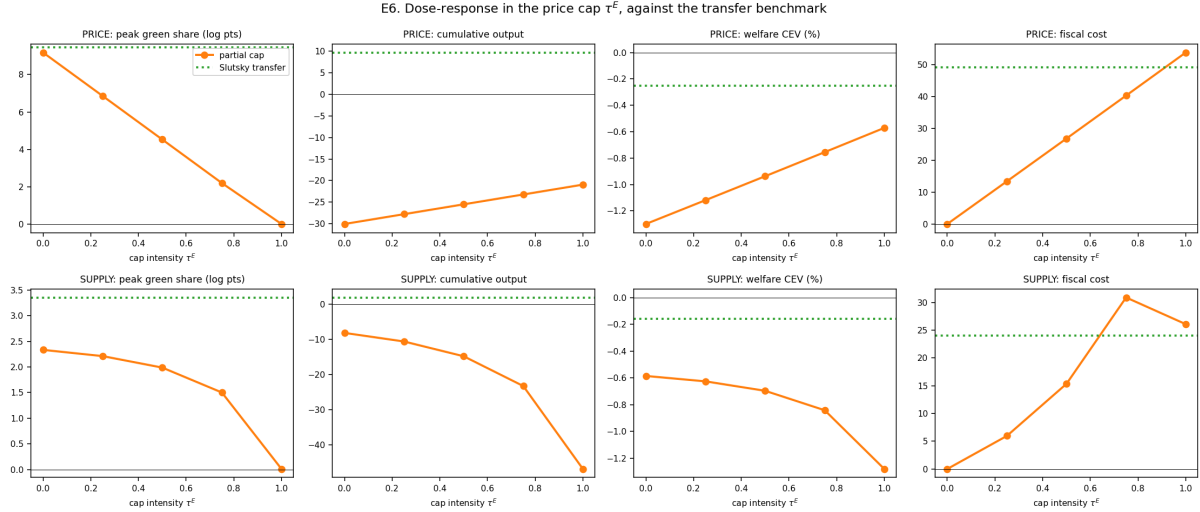


Figure 8: Dose-response in τ^E (orange) against the Slutsky transfer benchmark (dotted). Note the sign reversal of the welfare panel between the two shocks.

Private welfare and the transition are distinct. The CEV above is a *private* household-welfare measure: it values consumption and leisure along the transition and prices no climate or transition externality. Two results should therefore be kept separate. First, on private welfare alone the Slutsky transfer dominates the cap under *both* shocks (-0.25 vs. -0.57 under the price shock, -0.16 vs. -1.28 under the supply shock, Table 5); the case for the transfer over the cap does not rest on the transition. Second, and separately, the transfer preserves the green adoption margin that the cap eliminates (Section 4.3)—a positive statement about the energy transition, not a welfare ranking. We deliberately do not aggregate the two into a single social-welfare number: that would require a social value of adoption (a carbon price times avoided brown-energy use) that we do not model. A positive such value only reinforces the private ranking; it cannot reverse it here, since the transfer already dominates.

4.8 Distributional incidence

Scenario (price shock)	CEV mean	impatient	middle	patient
no policy	−1.30	−2.10	−1.65	−0.14
cap $\tau^E = 0.5$	−0.94	−1.45	−1.25	−0.11
cap $\tau^E = 1$	−0.57	−0.78	−0.85	−0.08
transfer	−0.25	−0.43	−0.50	+0.17

Table 6: CEV by discount-factor type (%). Impatient types hold little wealth and have high MPCs. Private household welfare only.

Without policy the impatient lose far more than the patient (-2.10 vs. -0.14 , roughly $15\times$). The patient remain the least-hurt group throughout the cap sweep. One clear distributional result: under the transfer, the **patient type’s CEV turns positive** ($+0.17\%$) while impatient and middle types remain negative. Indexed to pre-crisis energy use, the Slutsky compensation over-compensates low-MPC, low-energy-exposure households relative to their true loss—a symptom of the fiscal block’s lack of investment margin to crowd out the transfer (Section 6).

5 Relation to the literature

This section is written for the coauthors: it states, paper by paper, what our setup changes relative to the reference and what that change does to the results. Compact one-page summaries of each reference are in Appendix A.

5.1 Relative to Auclert et al. (2024)

Setup. We inherit the entire ARS aggregate economy—the CES energy nest, sticky import prices, the mutual fund, the energy suppliers with intertemporal arbitrage, the real-wage-stabilising wage Phillips curve, and the constant-real-rate monetary baseline—and change the household in four coupled ways. (i) ARS’s household is homothetic with a fixed energy technology: every household shares one CES basket and one CPI, and energy demand is $\alpha_E (P_E/P)^{-\eta_E} c$. We add a discrete brown/green durable state and a DC-EGM adoption choice, so the energy technology becomes an endogenous, household-level object. (ii) ARS has a single household energy price; we have two (P_B^E, P_G^E) and a type-specific cost-of-living index. (iii) ARS uses the CPI as unit of account; we use the domestic good, so that the resource constraint is invariant to how the CPI weights brown and green energy. (iv) We add two current-account terms absent in ARS—the switching cost as an import, the adopters’ energy saving as an import saving. The energy-supply closure is a generalisation of ARS: their baseline is our $\varepsilon^E = \infty$ price-taking case, and their coordinated world equilibrium (endogenous world price) is the two-country analogue of our ε^E -finite case.

Results. ARS’s central mechanism—low χ makes the energy price shock contractionary through the real-income channel—is preserved and operates in the background of every one of our experiments. Where we depart is that ARS’s fiscal ranking is entirely about output, inflation and inequality: in ARS the subsidy is a domestic “silver bullet” that stabilises all three, and its only drawback is a *cross-country* externality (coordinated subsidies spike the world price and become self-defeating). Our model adds an orthogonal cost of the same subsidy that ARS cannot see: capping P_B^E eliminates the green adoption margin (peak D^G from 9.18 to 0.01 pp), so the subsidy is not neutral for the transition. Two further connections are worth flagging to the coauthors. First, our supply-shock result—the cap is destructive under inelastic supply—is the *single-country* counterpart of ARS’s coordination result: in both, capping the price under an endogenous/scarce energy price prevents the necessary substitution and deepens the recession. Second, our monetary baseline being ARS’s constant-real-rate rule is what pins the real exchange rate, so a domestic transfer has no terms-of-trade effect; this is inherited, not assumed anew.

5.2 Relative to Bayer et al. (2026)

Setup. Bayer et al. is a two-country HANK monetary union (Germany one-third, rest of the union two-thirds), with a fixed nominal exchange rate, built to study *spillovers*. Our economy is a single small open economy with an exogenous foreign block, so by construction we cannot represent their central object—the cross-border, beggar-thy-neighbour transmission of a national subsidy. Three further setup differences matter. (i) Supply: Bayer fix the union-wide energy *quantity* and let a common price clear; this is exactly our ε^E -finite closure, so our supply-shock

experiments are the natural comparison. Our shock is smaller (a 10% cut for 6 quarters against their 20% cut for 6 quarters), chosen for first-order reliability. (ii) Supply side: Bayer has a rich medium-scale block—two assets (liquid bonds, illiquid capital), an investment margin with capacity utilisation, energy in production as well as consumption, and GHH preferences. We have a single asset, no capital, and energy in consumption only in the baseline. (iii) Heterogeneity: Bayer’s headline distributional mechanism is *within-country energy-share heterogeneity*—poor households have systematically higher energy shares (a high/low-intensity Markov type correlated with income), so they suffer larger real-income losses. We assume homothetic energy demand conditional on the durable type, so we cannot speak to their energy-exposure distributional results; our heterogeneity is over discount factors and over the adoption margin instead. Finally, their monetary policy is a standard Taylor rule, not the constant-real-rate rule; they show this matters, because under a Taylor rule the subsidy also suppresses *intertemporal* substitution through household-specific real rates.

Results. Under the supply closure we reproduce the qualitative core of Bayer et al.: the Slutsky/Hicks transfer welfare-dominates the price cap (our private CEV -0.16 vs. -1.28), and capping the price under a fixed quantity is inefficient because it prevents the substitution the scarcity requires—our “cap pushes towards Leontief” is their “subsidising under a supply contraction is inefficient.” What we add is the adoption margin, which Bayer et al. do not have: the same cap that is inefficient for output also switches off the transition, and the transfer that is welfare-superior also preserves it. What we *cannot* deliver is their spillover result: with no second country, we have nothing to say about whether one country’s subsidy harms its neighbours. This is the reason we are careful (Section 4.6, and the abstract) not to claim that a single parameter “reconciles” Bayer and Langot; we reproduce the domestic sign, not the cross-border mechanism.

5.3 Relative to [Langot et al. \(2026\)](#)

Setup. Langot et al. study France as a single economy inside the euro area, with an ECB Taylor rule down-weighted for France’s size. Two structural choices distinguish their model from ours. (i) They take the energy price as *exogenous* (fed from the government’s official forecasts), so France is a price taker facing a perfectly elastic supply at that price—this is our $\varepsilon^E = \infty$ closure, and their price-shock scenario is the right comparison. (ii) Their energy substitution elasticity is $\eta_E = 0.5$, five times ours, and they add an *incompressible* (subsistence) level of energy consumption, which delivers non-homotheticity: poor households have a larger incompressible share, hence a higher effective energy share and a lower price elasticity. This is their distributional mechanism, analogous in spirit to Bayer’s energy-share heterogeneity but generated by subsistence rather than by types. Our model buys energy substitution on a different margin: we keep the *intensive* elasticity low ($\eta_E = 0.1$, as in ARS) and obtain substitution from the *extensive* adoption margin instead. A third, methodological difference matters for how the results should be read: Langot et al. conduct an *ex-ante, conditional-forecast* evaluation—they back out the sequence of structural shocks that makes the HANK reproduce the French government’s published forecasts for output, inflation and debt, then run counterfactual policies holding those shocks fixed. Ours (like ARS and Bayer) is a clean MIT-shock impulse-response exercise, not a fit to a specific historical episode.

Results. Under the elastic (price) closure we agree with Langot’s headline: the cap cushions output ($\sum y = -20.9$ vs. -30.1 with no policy) and, on private welfare, tightening the cap raises the CEV ($-1.30 \rightarrow -0.57$)—consistent with their finding that the shield supports growth and curbs inflation. But two qualifications are important for the coauthors. First, Langot’s pro-shield result rests substantially on a *labour-cost* channel: by containing the retail energy price the shield prevents the price–wage spiral, holds down the cost of labour, and thereby supports employment; in their comparison the shield beats both wage indexation and a targeted transfer precisely on

this employment margin. Our wage block and our lack of an investment/employment margin mean we reproduce the *sign* of their result but not the full mechanism—hence we do not claim the Langot–Bayer contrast is “only” about the supply elasticity. Second, Langot et al. hold the energy technology fixed, so like ARS and Bayer they do not see the cap’s cost to the transition, which is our contribution.

6 Limitations

1. **The taste scale σ_ε is calibrated, not estimated.** ψ_g and σ_ε jointly hit $D_{ss}^G = 0.05$, but one level moment cannot separately identify both a switching cost and a taste scale: a second moment (e.g. the semi-elasticity of adoption to a purchase subsidy) is needed and would discipline the sign found in Section 4.1 along with the magnitudes throughout. The *ordering* of the cap-vs-transfer results does not depend on it, since it follows from the relative price alone.
2. **The switching cost is booked as an import by accounting necessity.** Under ARS’s $\mu_{ss} > 1$ with capitalised profits, booking $\psi_g D^{sw}$ as domestic demand creates a marginal dividend owned by nobody and breaks asset-market clearing; we therefore book it externally (Section 2.8). This convention is what makes the adoption channel *contractionary* under the price shock, since the cost lands in the periods household income is falling hardest (Section 4.1). The *ordering* of the policy results does not depend on it—it follows from the relative price alone—but the sign of the adoption channel’s output contribution under a price shock does, and should be read as conditional on this booking.
3. **Zero pass-through to the green price ($\rho_g = 0$)** means adopters are completely insulated from the fossil shock: an upper bound on the adoption channel. A partial pass-through calibrated to observed electricity–gas co-movement would be more defensible.
4. **No modelled green technology.** The adopters’ resource saving rests on the domestic/imported interpretation of Section 2.8 rather than on a green sector with its own capacity, cost curve and investment. We have implemented the minimal version—a competitive, zero-profit installer sector that books the switching cost as domestic value added rebated to households—and it clears to machine precision and flips the sign of the adoption channel (Section 4.2); what remains unmodelled is the green energy *supply* technology (capacity, cost curve, investment) and a distribution of installer income more realistic than the lump-sum rebate.
5. **No within-country energy-share heterogeneity.** Bayer et al. (2026)’s central distributional mechanism is that poor households have higher energy shares; we have homothetic energy demand conditional on the durable type, so we cannot speak to their distributional results.
6. **No investment margin.** Both Bayer et al. (2026) and Auclert et al. (2024) have richer supply sides. Its absence here likely makes the transfer look more expansionary, and the supply shock more inflationary, than it would be with capital to smooth the adjustment.

A Summaries of the three reference papers

These summaries are in our own words, for internal reference. They are deliberately compact—roughly one page each—and emphasise the features that matter for our comparison in Section 5.

A.1 Auclert et al. (2024), “Managing an Energy Shock”

Auclert et al. (2024) ask how an energy-importing economy should respond to an energy price shock, and whether household heterogeneity changes the answer. The model is a small open economy HANK built on the exchange-rate model of Auclert, Rognlie, Souchier and Straub (2021), extended with an energy good. There are three goods: a domestically produced home good, an imported foreign good, and imported energy. Households face idiosyncratic productivity risk, cannot insure it, and hold a single asset through a mutual fund; preferences are CRRA in a CES consumption basket that nests energy against a home/foreign bundle, with a low energy substitution elasticity ($\eta_E = 0.1$) and a total substitution index $\chi = 0.3$. Crucially, all households share the *same* basket and the *same* price index—the energy technology is fixed and homothetic. Wages are sticky with a real-wage-stabilisation motive; the monetary baseline is a rule that holds the real interest rate constant. Energy is supplied by price-taking foreign firms with an intertemporal extraction margin, so in the single-country equilibrium the world energy price is exogenous.

The analytical core is a decomposition of the output response into an expenditure-switching channel (positive) and a real-income channel (negative) scaled by the matrix of intertemporal MPCs. In the complete-markets representative-agent benchmark only expenditure switching survives, so the shock is expansionary; with realistic MPCs and low χ the real-income channel dominates and the shock is contractionary—their central result, and the one we inherit. A neutrality result holds at $\chi = 1$, where the two channels cancel. With real wage rigidity a wage–price spiral can appear, but the real wage still falls.

On policy, ARS study monetary tightening and three fiscal tools—energy subsidies, targeted transfers (indexed to counterfactual energy use), and untargeted transfers, all deficit-financed and repaid by a labour tax. For a country acting alone, subsidies are the most effective domestic stabiliser and tame inflation (they cap the price that drives the real-wage loss); transfers support demand but are more inflationary. The decisive twist is *cross-country*: monetary tightening has a positive externality (it lowers world energy demand and price), while subsidies have a large negative externality (coordinated subsidies make world energy demand nearly inelastic, spike the world price, and become self-defeating). Their policy recommendation is coordinated tightening plus transfers targeted to the poor, avoiding subsidies. They also show the impulse responses are not strongly nonlinear at empirically relevant shock sizes.

For our purposes the key features are: fixed homothetic energy technology, a single energy price and CPI, the constant-real-rate monetary rule, and the fact that the only “cost” of a subsidy in their world is a cross-country externality—there is no transition margin.

A.2 Bayer et al. (2026), “Hicks in HANK” / “Redistribution Within and Across Borders”

Bayer et al. (2026) evaluate the fiscal response to the European energy crisis in a two-country HANK model of a monetary union, calibrated with Germany as “Home” (one-third of the union) and the rest of the area as “Foreign” (calibrated to Italy). The model is a medium-scale, two-asset HANK: households hold a liquid union-wide bond and illiquid capital, face idiosyncratic income risk, and have GHH preferences; there is an investment margin with capacity utilisation. Energy enters both production (a CES bundle with the capital–labour aggregate, elasticity 0.2) and household consumption (a CES bundle with the physical good, elasticity 0.1). A distinctive and, for them, central feature is within-country heterogeneity in the *energy share*: households are of high- or low-energy-intensity type, following a persistent Markov process correlated with income, so that poorer households spend a larger share on energy and suffer larger real-income losses. Monetary policy is a standard union-wide Taylor rule; each national fiscal authority taxes labour and profits and stabilises its debt.

The defining assumption is that the total union energy supply is *inelastic*: import capacity

is fixed in the short run, so a fixed quantity is set exogenously and a common price clears the market. The crisis is a 20% cut in supply for six quarters, in response to which the energy price rises roughly fivefold, inflation rises about four points, and output falls around one percent, with consumption falling most for poor, high-energy households.

They compare two policies, each implemented in Home only so that spillovers can be measured: a subsidy that caps the retail energy price at its pre-crisis level, and a Hicks/Slutsky transfer indexed to pre-crisis energy consumption (to households and firms). The subsidy is the more effective domestic stabiliser but is fiscally expensive and *beggar-thy-neighbour*: by holding Home’s energy demand up it pushes the common price higher, amplifying the crisis in Foreign. The transfer stabilises less at Home but generates essentially no spillover and is cheaper. On welfare (consumption-equivalent variation), the subsidy *reduces* Home welfare—subsidising energy under a supply contraction is inefficient because it suppresses both intratemporal and intertemporal substitution—while the transfer *raises* Home welfare, its cost being mostly the distortion of the taxes that finance it. Their headline contrast with ARS (subsidy more stabilising than transfer for output) they attribute to the Taylor rule and the two-country structure, versus ARS’s constant-real-rate rule.

For our purposes the key features are: fixed union-wide quantity (our inelastic closure), the two-country spillover mechanism that we cannot represent, the within-country energy-share heterogeneity that we do not have, and the substitution-preserving property of the transfer, which we share.

A.3 Langot et al. (2026), “The Macroeconomic and Redistributive Effects of the French Tariff Shield”

Langot et al. (2026) evaluate the French *bouclier tarifaire*—the 2022–23 cap on gas and electricity prices—in a HANK model of France as a single economy within the euro area. The model extends Auclert, Rognlie, Souchier and Straub with energy in both household consumption and production. Households face idiosyncratic productivity risk and hold a single asset; preferences are CRRA with a CES energy nest, and the energy substitution elasticity is $\eta_E = 0.5$, markedly higher than in ARS or Bayer. A distinctive feature is an *incompressible* (subsistence) level of energy consumption: only energy above the subsistence level yields utility. This generates non-homotheticity without non-homothetic preferences—poorer households devote a larger share to incompressible energy and have a lower price elasticity of energy demand, which is the source of the distributional results. The ECB Taylor rule is down-weighted to reflect France’s small weight in euro-area inflation. In the policy simulations the energy price is taken as exogenous (from the government’s forecasts), so France is effectively a price taker facing perfectly elastic supply.

The methodology is distinctive and worth flagging: rather than clean MIT-shock impulse responses, the paper performs an *ex-ante, conditional-forecast* evaluation. Using the sequence-space Jacobian, the authors back out the sequence of structural shocks (energy price, government spending, transfers, plus latent demand, markup and fiscal shocks) that makes the model reproduce the French government’s official published forecasts for output, inflation and the debt ratio through 2027. Policies are then evaluated as counterfactuals that hold this shock sequence fixed and change one instrument.

The main finding is that the tariff shield supports growth (about 1.9% average over 2022–23 against 1.0% without it) and curbs inflation (5.6% against 7.2%), at a fiscal cost near 2% of GDP per year and a modest rise in the debt ratio, while containing—though not eliminating—the rise in consumption inequality even though it is untargeted. The mechanism behind its effectiveness is a *labour-cost* channel: by holding down the retail energy price the shield prevents the price–wage spiral, contains the cost of labour, and thereby supports employment and output. On this macro criterion the shield beats both a faster wage indexation (more inflationary, worse for output) and a targeted transfer to finance incompressible energy consumption; the targeted transfer, however, reduces inequality by more.

For our purposes the key features are: exogenous/elastic energy price (our price-taking closure), the labour-cost channel behind the pro-shield result (which our wage block only partly captures), the higher intensive substitution elasticity, and again a fixed energy technology with no transition margin.

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